

Homework 1

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Problem 1

(a)

First I scale u by 2 and subtract v entry by entry, then dot the result with w :

$$2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = (3)(-4) + (-6)(2) + (8)(6) = -12 - 12 + 48 = 24$$

[ANSWER: 24]

(b)

Halving w first gives $(1/2)w = [-2, 1, 3]^T$, and adding that to v gives

$$v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T.$$

The length is the square root of the sum of the squares of the entries:

$$\|v + (1/2)w\| = \sqrt{(-1)^2 + 5^2 + 1^2} = \sqrt{1 + 25 + 1} = \sqrt{27}$$

Since $27 = 9 \cdot 3$, this simplifies to $3\sqrt{3}$, which is about 5.20.

[ANSWER: $3\sqrt{3} \approx 5.20$]

(c)

$\cos \theta = (u \cdot w) / (\|u\| \|w\|)$, so I need the dot product and both lengths first.

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8 / (2 \cdot 14) = 8/28 = 2/7$$

The inverse cosine of $2/7$ in degrees is $73.398\dots^\circ$, which I round to two decimals.

[ANSWER: $\theta \approx 73.40^\circ$]

Problem 2

(a)

AB is undefined, because A is 3×3 and B is 2×3 , so the 3 columns of A do not match the

2 rows of B . BA is defined (2×3 times 3×3) and the product is 2×3 .

$$\text{row 1} = [1(2) + 0(1) + (-2)(0), 1(-1) + 0(3) + (-2)(4), 1(0) + 0(-2) + (-2)(1)] = [2, -9, -2]$$

$$\text{row 2} = [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

[ANSWER: AB undefined; $BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$]

(b)

Av is defined (3×3 times 3×1) and the result is another 3×1 column:

$$Av = [2(1) + (-1)(-2) + 0(3), 1(1) + 3(-2) + (-2)(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$$

VA is undefined, because v is 3×1 and so it has only 1 column, while A has 3 rows.

[ANSWER: $Av = [4, -11, -5]^T$; VA undefined]

(c)

Aw is undefined: A has 3 columns but w is 1×3 , so it has only 1 row, and $3 \neq 1$.

WA is defined (1×3 times 3×3) and the answer is a 1×3 row vector:

$$WA = [2(2) + 0(1) + (-1)(0), 2(-1) + 0(3) + (-1)(4), 2(0) + 0(-2) + (-1)(1)] = [4, -6, -1]$$

[ANSWER: Aw undefined; $WA = [4 \ -6 \ -1]$]

(d)

Both exist but have different shapes: vw is 3×1 times 1×3 , so 3×3 , and wv is 1×1 .

$$vw = \begin{bmatrix} 1(2) & 1(0) & 1(-1) \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -2(2) & -2(0) & -2(-1) \end{bmatrix} = \begin{bmatrix} -4 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3(2) & 3(0) & 3(-1) \end{bmatrix} \begin{bmatrix} 6 & 0 & -3 \end{bmatrix}$$

$$wv = 2(1) + 0(-2) + (-1)(3) = 2 - 3 = -1, \text{ which is the dot product of } w \text{ and } v.$$

[ANSWER: vw as above; $wv = [-1]$]

(e)

A is square, so $A^2 = AA$ is defined and is again 3×3 . Multiplying row by column:

$$\text{row 1} = \begin{bmatrix} 2(2) + (-1)(1) + 0(0), & 2(-1) + (-1)(3) + 0(4), & 2(0) + (-1)(-2) + 0(1) \end{bmatrix} = \begin{bmatrix} 3, & -5, & 2 \end{bmatrix}$$

$$\text{row 2} = \begin{bmatrix} 1(2) + 3(1) + (-2)(0), & 1(-1) + 3(3) + (-2)(4), & 1(0) + 3(-2) + (-2)(1) \end{bmatrix} = \begin{bmatrix} 5, & 0, & -8 \end{bmatrix}$$

$$\text{row 3} = \begin{bmatrix} 0(2) + 4(1) + 1(0), & 0(-1) + 4(3) + 1(4), & 0(0) + 4(-2) + 1(1) \end{bmatrix} = \begin{bmatrix} 4, & 16, & -7 \end{bmatrix}$$

[ANSWER: $A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$]

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & -7 \end{bmatrix}$$

(f)

B^2 is undefined: squaring needs the columns to match the rows, but B is 2×3 , so BB

would require $3 = 2$.

[ANSWER: B^2 undefined (B is not square)]

Problem 3

(a)

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$\text{so } x_1 = 2, x_2 = -1, x_3 = 1$$

$$\text{[ANSWER: } x = [2, -1, 1]^T \text{]}$$

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -7 & | & 8 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

The bottom row is all zeros, so it tells me nothing and I drop it. Taking $z = 0$, the first row gives $x = 8$ and the second gives $y = -2$, so the system has one solution.

[ANSWER: $x = 8, y = -2, z = 0$]

Problem 4

(a)

Let x_1, x_2, x_3 be the masses in kilograms of Alloy 1, Alloy 2, Alloy 3 in the blend. The finished 100 kg batch is 36% copper, 44% zinc, 20% nickel, so it holds 36 kg of copper, 44 kg of zinc, 20 kg of nickel. No metal is lost, so for each metal the mass going in equals the mass coming out:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(b)

I multiplied every equation by 10 first so I wouldn't have to row reduce decimals:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$R_1 \rightarrow R_1 - 2R_3: [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_3 \rightarrow R_3 - 2R_1: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 10 \mid 200]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

Back substituting: $x_3 = 20$, then $x_2 = 80 - 2(20) = 40$, then $x_1 = -40 + 4(20) = 40$.

Checking against the copper equation, $0.5(40) + 0.2(40) + 0.4(20) = 20 + 8 + 8 = 36$.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, and 20 kg of Alloy 3]

Problem 5

(a)

$$\begin{array}{ccccc} [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 0] \\ [2 \ 4 \ 3] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] \\ [3 \ 6 \ 4] & [3 \ 6 \ 4] & [0 \ 0 \ 1] & [0 \ 0 \ 0] & [0 \ 0 \ 0] \end{array}$$

[ANSWER: $\text{RREF}(A) = [1 \ 2 \ 0; 0 \ 0 \ 1; 0 \ 0 \ 0]$]

(b)

$$\text{rank}(A) = 2$$

(c)

Zero solutions is possible. Running the same operations on $[A \mid b]$ turns the third row

into $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$. Whenever $b_3 \neq b_1 + b_2$, for instance $b = [0, 0, 1]^T$, that row reads $0 = 1$, which is false, so no x can satisfy the system.

Exactly one solution is impossible. Column 2 never becomes a pivot column, so x_2 is always a free variable once the system is consistent, and a free variable means more than one solution. In rank terms, $\text{rank}(A) = 2 < 3$ unknowns, so uniqueness can't occur.

Infinitely many solutions is possible. If $b_3 = b_1 + b_2$, say $b = [1, 3, 4]^T$ or $b = 0$,

the last row reads $0 = 0$ and the system is consistent, so the free variable x_2 gives an entire line of solutions.

Problem 6

(a)

These are dependent because there are three vectors and v_3 has a zero entry, so they can't fill $\mathbb{R}^3$.

[ANSWER: dependent; span is a plane]

(b)

Three vectors in $\mathbb{R}^3$ make a square matrix, so I can test independence by determinant:

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

Expanding along the first row with cofactors,

$$\det = 1(1 \cdot 1 - 0 \cdot 1) - 0(1 \cdot 1 - 0 \cdot 0) + 1(1 \cdot 1 - 1 \cdot 0) = 1 - 0 + 1 = 2.$$

The determinant is $2 \neq 0$, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ forces $c_1 = c_2 = c_3 = 0$ and the

vectors are independent. Three independent vectors in $\mathbb{R}^3$ have to span all of $\mathbb{R}^3$.

[ANSWER: independent; span is all of $\mathbb{R}^3$]